

User Manual of the Interactive Space Radiation Software

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1 Overview of the Application

This software has been developed utilizing the *Tkinter* graphics library of python3 and is intended exclusively for **educational purposes**. Various simplifications are incorporated into the code to enhance understanding and usability (for further details, refer to Sect. 3). Users have the flexibility to start the application either through a jupyter notebook or by executing it directly with python3 (additional information can be found in Sect. 2).

The main aim of this application is to explore the effects space radiation has on satellites and discover methods to safeguard them. In order to achieve this, the application enables users to

specify the path a satellite takes in orbit and to observe the radiation levels it encounters from the Sun, as well as from the Van Allen Belts (for further information, refer to Sect. 3). Additionally, users have the option to adjust the satellite's shielding parameters to assess enhancements in its durability and the subsequent extension of the mission's lifespan.

Upon execution, the program initiates a new window that displays six panels, as depicted in Fig. 1. The panels with red labels (1, 2, 3, and 5) serve as plot panels, showcasing diverse data, while the blue-labeled panels (4 and 6) function as configuration panels. In these configuration panels, users have the ability to adjust certain parameters through sliders or input fields. Specifically, Panel 4 contains all the satellite's orbital parameters, which are visually represented in Panel 1. This panel illustrates the Earth as a blue sphere, accompanied by the Van Allen belts depicted in red and green, and the Southern Atlantic Anomaly (SAA) highlighted in purple.

Additionally, Panel 4 includes an additional parameter, specifically called "Solar Activity Phase." This parameter, true to its name, determines the phase of solar activity within its typical 11-year cycle. The solar activity's impact on flux is illustrated in Panel 5, where the Sun's resulting flux (expressed in $W.m^{-2}$) is depicted as a function of this phase.

Based on the orbital parameters defined by the user, the code is designed to calculate the amount of radiation absorbed (in the absence of any shielding) per hour as the satellite travels along its orbit. The results of these calculations are displayed in panel 2. In every orbit, a satellite absorbs solar radiation. Moreover, its trajectory may result in the absorption of extra radiation when it traverses the Van Allen Belts.

Panel 6 provides crucial details, such as the Total Ionizing Dose (TID) and the Displacement Damage (DD) encountered during each orbit, and it enables users to customize the spacecraft's shielding accordingly. Additionally, this panel offers insights into the overall mass of the satellite, inclusive of the shielding materials, and highlights the additional expenses incurred for launching, attributable to the implemented shielding.

Ultimately, Panel 5 illustrates the trend of the TID and DD as they vary over time, continuing this observation until one of these metrics attains its designated threshold, which is typically regarded as the **End of Mission**. Further specifics on the computation methods can be found in Sect. 3.5.

The process of calculating the Total Ionizing Dose (TID) and Displacement Damage (DD) over time, as depicted in Panel 3, involves an integration method, which might require a bit of time to complete, often amounting to several seconds. To keep users informed about the status of this computation, a progress bar is displayed at the bottom of the window, visually indicating the advancement of the process.

2 How to Use

2.1 Requirement

- python version 3 (tested with python3.10.12)

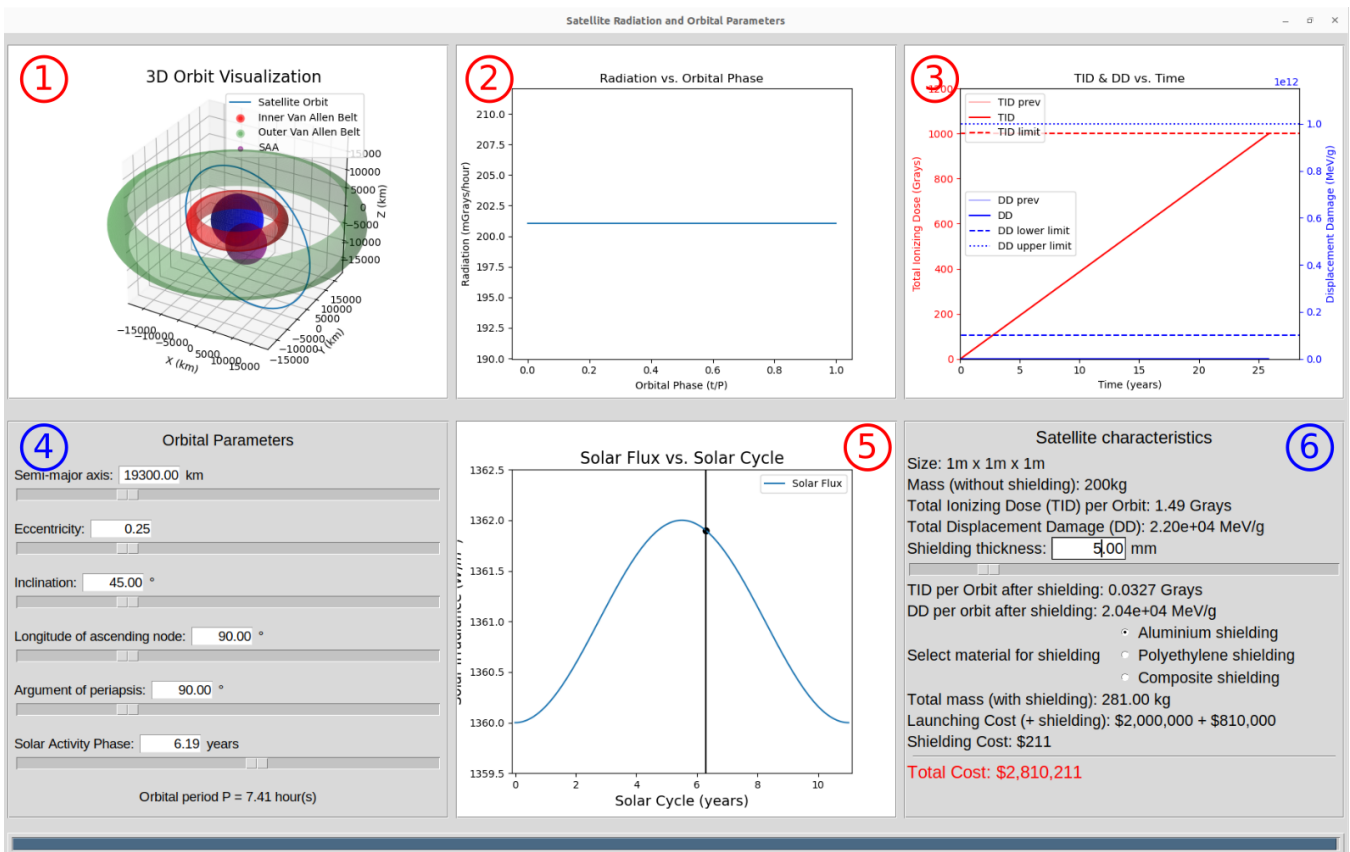


Figure 1: Screenshot of the Application window with enumerated panels for references.

- matplotlib library
- tkinter library
- numpy library
- scipy library
- math library
- (optional) jupyter notebook or jupyter lab: required if the App should be launched from the notebook (.ipynb file)

2.2 Get the App

To get the software (both the notebook and the python script), you should clone the git repository at https://github.com/naimar-pro/Space_radiation.git

```
git clone git@github.com:naimar-pro/Space_radiation.git
```

2.3 Launching the application

2.3.1 From a Terminal

To launch the application from a terminal, simply use the following command:

```
python3 interactive_Space_radiation.py
```

Do not close the terminal during the use of the application, it will close the App. A proper way to close the application is to click the classical `close` button located in the top right part in Fig. 1. **Note that the position of this button may depend on your Operating System.**

2.3.2 From a notebook

To launch the App from the notebook, the first step is to open the notebook itself:

```
jupyter-notebook interactive_Space_radiation.ipynb
```

or

```
jupyter lab interactive_Space_radiation.ipynb
```

To launch the App click the button **restart the kernel and run all cells** (button labeled 2 in blue in Fig. 2). After a few seconds, a new window will appear.

To close the App, come back to the notebook and click the button **restart the kernel** (button labeled 1 in red in Fig. 2).

3 Approximations and hypothesis

3.1 Orbit

The trajectory is considered to be an elliptical path as described by Kepler's Laws, and it is assumed that there is no drag force acting on it. Consequently, the parameters defining the orbit remain constant over time, ensuring that the orbital path does not change indefinitely.

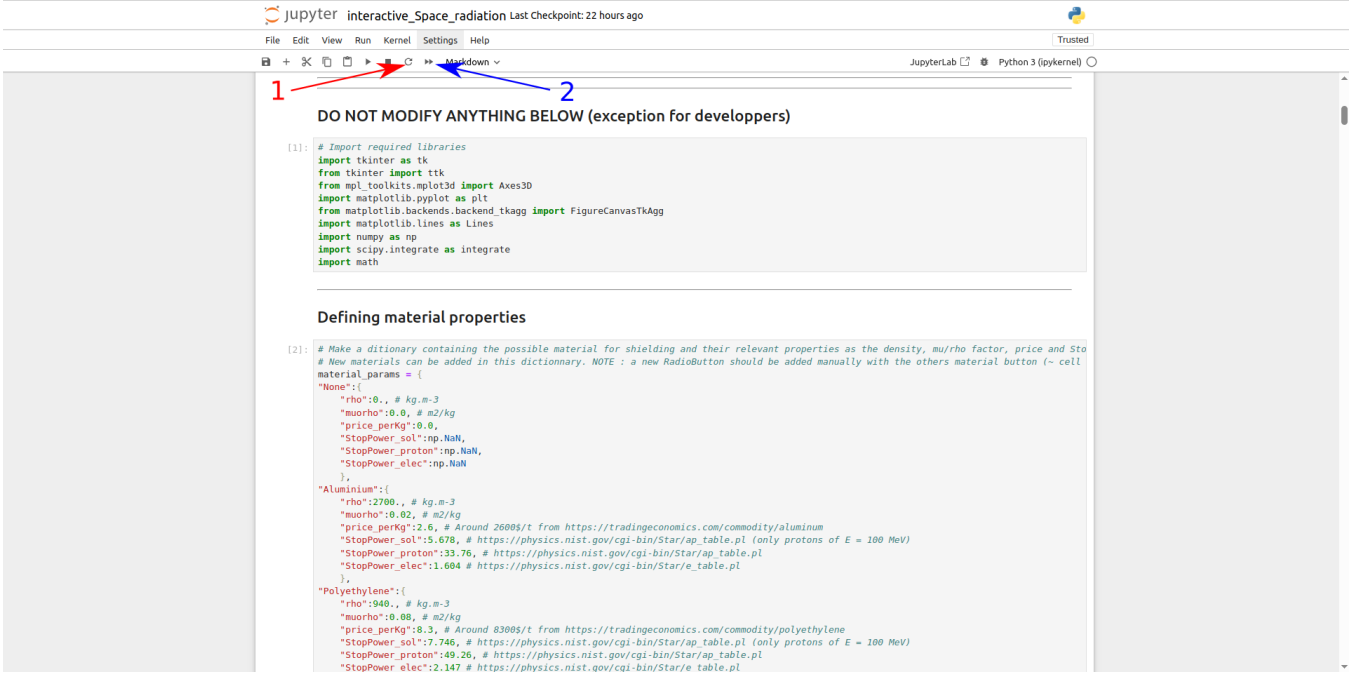


Figure 2: Screenshot of the notebook in a web browser.

3.2 Solar radiation

3.2.1 Solar Flux

The sun undergoes an 11-year cycle during which the solar flux received by Earth fluctuates. This modulation is often described through a basic sinusoidal function. Nonetheless, the variation in solar flux is minimal, characterized by an average flux of approximately 1361 W.m^{-2} and an amplitude of fluctuation around $\sim 2 \text{ W.m}^{-2}$.

3.2.2 Ionizing Dose

A limited portion of the solar radiation, specifically extreme ultraviolet (eUV) rays, X-rays, and γ -rays, actually influences satellites. In theory, the quantity of these eUV rays, X-rays, and γ -rays that the Sun releases varies throughout its cycle. However, for the sake of simplicity in our analysis, we consider the fraction of this ionizing radiation to remain a fixed value, set at 10^{-6} .

To transform this measurement of radiation into milligray per hour, we can apply the following equation:

$$D_0 = \frac{P \cdot t \cdot \mu}{\rho} \quad (1)$$

In this equation, D represents the radiation dose absorbed in terms of Gray. The variable P denotes the solar flux measured in watts per square meter (W.m^{-2}). The variable t stands for the time of exposure in seconds. Meanwhile, μ/ρ refers to the mass attenuation coefficient expressed in square meters per kilogram (m^2/kg) and is a property specific to the substance that has absorbed the radiation.

Typically, in an unshielded satellite core, a significant portion of the material is comprised of Aluminum, which constitutes approximately 50% of the structure. In addition, composite materials are employed for both structural components and panels, making up around 30% of the core

composition. The remaining 20% is devoted to the electronic circuit. By utilizing the respective values of mass attenuation coefficient over density, μ/ρ , for each material, and their corresponding percentages, one can calculate an average value of the mass attenuation coefficient over density, $\mu_{\text{avg}}/\rho_{\text{avg}}$, equating to $0.041 \text{ m}^2/\text{kg}$. To determine the radiation dose received per hour, one can use a time duration of $t = 3600 \text{ s}$, which subsequently allows for the integration of this dosage.

3.2.3 Displacement Damage

The Sun not only emits electromagnetic radiation but also releases energetic particles in a phenomenon known as the solar wind. The energy levels and the intensity of this solar wind are significantly influenced by solar activity. It is important to note that only the most energetic particles, those typically with energies greater than 100 MeV, can penetrate the Earth's magnetosphere without being deflected. Particles possessing lower energies are instead confined within the Van Allen Belts—please refer to the section below on Van Allen Belts for more details. These particles pose a risk of damage to satellites, particularly affecting their electronic components.

For simplicity in our analysis, let us assume that all solar wind particles are protons and each possesses an energy of 100 MeV. According to Sect. 4.3 from the document accessible at [this link](#), we can deduce the average flux of these particles, which is approximately 82.4 particles per second. However, it is crucial to consider that this flux is influenced by modulations related to solar activity.

Thus, the flux of solar particles as it varies over the years can be expressed in the following manner:

$$F_{\odot > 100 \text{ MeV}}(\text{year}) = 82.4 \times \left(1.1 + \sin \left(\frac{2\pi(\text{year} - 5.5/2)}{11} \right) \right). \quad (2)$$

According to this mathematical expression, the maximum flux occurs when the year is equal to 5.5 modulo 11 years.

The Displacement Damage caused as a result of the solar wind is subsequently calculated through the equation:

$$DD_{\odot > 100 \text{ MeV}}^0 = F_{\odot > 100 \text{ MeV}}(\text{year}) \times P \times NIEL_p \quad (3)$$

In this equation, P denotes the orbital period measured in seconds, and the parameter $NIEL_p$ represents the Non-Ionizing Energy Loss coefficient for protons. This coefficient is valued at $5 \times 10^{-3} \text{ MeV.cm}^2/\text{g}$.

3.3 Van Allen Belts

3.3.1 Geometry

The Van Allen belts, known for their intricate and dynamic geometry, undergo variations as time progresses. These radiation belts can be conceptually understood by imagining them as dipoles whose axes are tilted relative to Earth's rotational axis. The composition of the Van Allen belts consists of an inner and an outer region. The inner belt is mainly made up of energetic protons, while the outer belt contains high-energy electrons. For simplification purposes and as an initial approximation in our model, the two Van Allen belts—namely the inner and outer belts—are represented more straightforwardly in the computational code as two toroidal structures. The inner belt is depicted to exist between altitudes of 1,000 km and 6,000 km, whereas the outer belt spans altitudes ranging from 15,000 km to 25,000 km. In the graphical visualization within Panel 1, these belts are visually distinguished by color: the inner belt appears in red and the outer belt in green. It is important to note, however, that the inclination angle between the magnetic dipole axis and the Earth's rotational axis is not incorporated into this model representation.

3.3.2 Total Ionizing Dose

To determine the amount of ionizing dose experienced within the radiation belts, we employ an analytical expression that varies with the radial distance r from Earth's center. This formula is crucial for understanding the dose dependency based on spatial position:

$$D_0(r) = 60 \left(\frac{r}{25000} \right)^2 \text{ mGray/hour} \quad (4)$$

where r is measured in kilometers. While this offers a significant simplification, it is important to note that the total ionizing dose (TID) is significantly less impacted by the radiation belts in comparison to the overall effect they have on the Displacement Discharge (DD).

3.3.3 Displacement Damage

The initial task involves calculating the fluence of energetic particles residing within the two distinct radiation belts, expressed in units of particles per square centimeter per second. This calculation can be formulated as shown below:

$$F_i = \rho_i v_i \quad (5)$$

Here, the subscript i is utilized to specify the particle category, with protons relating to the inner belt and electrons corresponding to the outer belt. It is important to note that we operate under the assumption that the particle density remains uniform across their respective belts. According to Mauk et al. (2012), the particle density within the inner belt is approximately 2×10^5 particles per cubic centimeter for protons, while in the outer belt, the density for electrons is noted to be 10^6 particles per cubic centimeter (see Fig. 1 of the cited work).

To determine the velocities of these particles, we apply the relativistic relation for their kinetic energy, denoted as E_k , which is given by:

$$E_k^i = (\gamma - 1)m_i c^2 = \left(\frac{1}{1 - \frac{v_i^2}{c^2}} - 1 \right) m_i c^2 \quad (6)$$

This equation can be rearranged to express the velocity as follows:

$$v_i = c \sqrt{1 - \left(\frac{m_i c^2}{E_k^i + m_i c^2} \right)^2} \quad (7)$$

Analogous to the situation with the solar wind, we maintain an assumption that all particles possess a singular energy level of 10 MeV for the protons located in the inner belt, and 0.5 MeV for the specific case of the electrons, as illustrated in Fig. 1 of Mauk et al. 2012.

The DD per particle type is thus

$$DD_i^0 = F_i \cdot T_i \cdot NIEL_i \quad (8)$$

with T_i is the time spent in the given Van Allen belt.

3.4 Shielding

To mitigate the impact of space radiation and energetic particles on spacecraft—specifically in terms of Total Ionizing Dose (TID) and Displacement Damage (DD)—we can employ materials with high density. These materials are capable of absorbing a substantial portion of the incoming radiation and energetic particles, subsequently reducing the exposure endured by the satellite’s components. We will focus our considerations on three particular materials: aluminum, polyethylene, and a composite material that incorporates carbon fiber.

3.4.1 Attenuation coefficient for TID

Starting from basic principles, the attenuation of Total Ionizing Dose (TID) can typically be described as an exponential decrease, where the reduction in TID is dependent on the material’s thickness. Important material properties to consider in this context include the density ρ and the ratio of the attenuation coefficient to density, μ/ρ . Nonetheless, experimental results documented in the file named *Introduction_To_Radiation_Shielding.pdf* (located in the Biblio folder) reveal an unexpected behavior. Consequently, we have modified the formula for calculating the attenuation coefficient to better align with the empirical data presented in *Introduction_To_Radiation_Shielding.pdf*. The adapted expression is given by:

$$D(d) = D_0 e^{-\frac{\mu}{\rho} \rho d^2} \quad (9)$$

Here, d represents the thickness of the shielding material, while D_0 denotes the ionizing dose measured before the presence of the shielding. It is important to emphasize that this equation is crafted as an approximation specifically aimed at replicating experimental data that was initially gathered for Aluminum. This approximation has subsequently been extended for application to materials such as Polyethylene and Composite.

3.4.2 Attenuation coefficient for DD

To calculate the extent to which the displacement damage (DD) is reduced by the presence of shielding—essentially determining how much a material absorbs incoming energetic particles—we utilize the concept known as **Stopping Power** for the materials involved. The analysis involves segregating the computation of the attenuation coefficient into three distinct particle types. These are: protons originating from the solar wind with an energy level of $E_{\odot} = 100$ MeV; protons located within the inner Van Allen belt identified by an energy level of $E_p = 10$ MeV; and electrons that are found in the outer Van Allen belt possessing an energy of $E_{e-} = 0.5$ MeV. For each type of particles, we determine the stopping length denoted as λ_i for the material and compute it using the formula:

$$\lambda_i = \frac{E_i}{SP_i \cdot \rho}, \quad (10)$$

Here, the subscript i denotes the specific type of particles: be it solar protons, Van Allen belt protons, or Van Allen belt electrons. The term SP_i represents the stopping power measured in MeV.cm²/g, which can be obtained from well-known resources such as the [ESTAR](#) and [PSTAR](#) databases. According to the guidelines outlined in the [Total Dose Modeling of the Transit Through the Van Allen Belts](#) document provided by NASA, the DD affecting the internal portions of a satellite relative to the shielding’s thickness is typically represented by power-law relationships with an empirically determined index of -2.19 . For normalization, we require that the displacement

damage coefficient is equal to 1 when there is no shielding thickness. Thus, we calculate the DD attenuation coefficient using the expression:

$$DD_i(d) = DD_i^0 \times \begin{cases} e^{-d/\lambda} & \text{if } d \leq \lambda, \\ e^{-1} \left(\frac{d}{\lambda}\right)^{-2.19} & \text{otherwise.} \end{cases} \quad (11)$$

By summing the individual DD_i values computed for the three particle types, we ultimately derive the total Displacement Damage.

3.5 Integration of TID and DD

To construct the plot shown in Panel 3, which depicts both the Total Ionizing Dose (TID) and Displacement Damage (DD) over time until one value hits its threshold (1000 Grays for TID and 10^{12} for DD), we initially calculate the TID and DD for each orbit. This calculation is performed with minimal solar activity to ensure that we obtain the smallest possible values. An estimate is made to determine how many orbits are required to reach the specified limit, and we choose the smallest of these numbers, ensuring a baseline of at least two orbits, as the quantity of orbits for integration. Should this figure exceed 500, we proceed by integrating over 500 increments with multiple orbits per increment. Conversely, if the number of orbits does not exceed 500, the number of increments matches the orbit count. This approach helps to minimize the computational duration required to generate the plot. The end time is derived by multiplying the total number of orbits by the orbital period. At each increment, we calculate the TID and DD per orbit at the corresponding time, considering the influence of the solar activity phase. We then compute and apply the relevant attenuation coefficients and aggregate the resulting values to the previously integrated totals, initiating from zero. In this manner, we are able to depict the cumulative TID and DD as functions of time.